

SAJIB RAFI HOSSAIN

929-472-5332 — sajibrafihoossain@gmail.com — [linkedin.com/in/sajib-rafi-hossain](https://www.linkedin.com/in/sajib-rafi-hossain) — github.com/rafihooss — rafihooss.github.io/Landing-Page/

Education

Queens College, City University of New York

Bachelor's in Computer Science

Expected May 2027

Queens, New York

Technical Skills

Languages: Python · JavaScript · TypeScript · Java · C++ · HTML · CSS

Technologies/Frameworks: React.js · Next.js · Numpy · pandas · Matplotlib · scikit-learn · Tailwind CSS · Node.js · Express.js · Drizzle ORM · PostgreSQL · SQLite · DynamoDB · AWS · Supabase · Git/Github · Jupyter Notebook

Experience

Break Through Tech

March 2026 - Present

AI/ML Fellow

Remote

- Applied CRISP-DM framework to translate business questions into machine learning problems across multiple datasets
- Built and analyzed datasets using NumPy, pandas, & Matplotlib DataFrame creation, data loading, and visualization
- Cleaned and prepared data for modeling by identifying features and labels, handling missing values, winsorizing outliers, and applying one-hot encoding

CUNY Tech Prep

July 2025 - Present

Software Engineer Fellow

Remote

- Selected for a competitive, year-long fellowship for top CUNY computer science students, focusing on full-stack web development, cloud technologies, and professional software engineering practices
- Practicing system design by translating requirements into scalable architectures, defining APIs, modeling data, and evaluating tradeoffs (performance, cost, reliability) through diagrams, written docs, and iterative design reviews
- Building full-stack applications with React/Next.js (TypeScript), Tailwind CSS, Node.js/Express, PostgreSQL, DynamoDB, Supabase, and AWS, supported by CI/CD pipelines with GitHub Actions
- Implementing engineering best practices, including minimum **70%** test coverage, Lambda API development, infrastructure testing with LocalStack, and robust documentation (ADRs, READMEs, API docs)

CodePath

September 2023 – April 2024

Web Developer Fellow

Remote

- Strengthened skills in HTML, CSS, JavaScript, and React through projects, including API integration and & parsing
- Collaborated in team-based coding sessions, improving project delivery through code reviews and pair programming

Projects

ClearSource | *Next.js, TypeScript, PostgreSQL, Prisma, RAG*

May 2026 - Present

- Building ClearSource, a vegan product verification app that uses a retrieval-first RAG workflow to compare submitted product data against a verified knowledge base and return evidence-backed match explanations
- Creating a human-in-the-loop verification pipeline that flags uncertain ingredients, drafts manufacturer inquiries, supports admin-reviewed final decision, and future embedding/pgvector-based retrieval

GreenBites | *TypeScript, React, Next.js, Supabase*

December 2025

- Developed a full-stack nutrition and sustainability tracker that lets users search foods via **Calorie Ninjas API**, adjust serving sizes, and track daily macros, calories, and carbon footprint with interactive dashboards
- Designed **PostgreSQL** database schema on Supabase with **Drizzle ORM**, protected routes with Clerk authentication, resolved production deployment issues, and fixed persistence issues so tracking remains accurate across sessions

CUNY GradeLens | *JavaScript, React, Node.js, Express.js, OpenAI API, SQLite, Tailwind CSS*

August 2025

- Built an AI-powered platform during the CTP Hackathon as part of a team of 3 that analyzes grade distributions with GPT-4, enabling **natural language queries** to deliver insights on professors and course difficulty
- Developed a chatbot with conversation history, fuzzy name matching algorithms, and data visualizations (**Chart.js**), robust **CSV data ingestion**, regex-based entity extraction, and multi-level caching to optimize API costs by **60%**

Leadership / Community Impact

STEM Instructor

July 2025 - August 2025

New York Edge

- Led environmental science and STEM lessons (sustainability, renewable energy, ecosystems, etc.) to **150+** grade 3–5 students; adapted instruction to diverse learners and fostered engagement through interactive, hands-on activities.